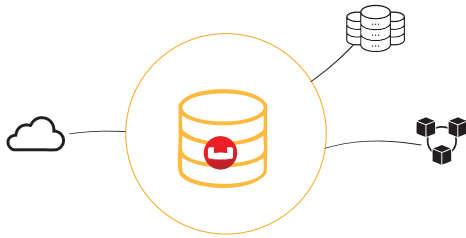


Couchbase: The Distributed Open Source Database

Couchbase boasts of a range of modern features, including memory-first architecture and a powerful query engine. Here's a short introduction to what it offers, with a quick guide on how to install it using Docker.



Couchbase is an award-winning, distributed NoSQL cloud database that delivers excellent versatility, performance, scalability, and financial value for all your cloud, mobile, on-premises, hybrid, distributed cloud, and computing applications.

It is a modern distributed document database that incorporates a built-in managed cache to enable memory-first architecture with all the desired capabilities of a relational database, and more. It exposes a scale-out, key-value store with managed cache for sub-millisecond data operations, purpose-built indexes for efficient queries, and a powerful query engine for executing SQL-like queries.

The services provided by Couchbase are as follows.

- **Data:** Supports the storage, setting, and retrieval of data items specified by key.
- **Query:** Parses queries specified in the N1QL query-language, executes the queries, and returns results. The Query service interacts with both the Data and Index services.
- **Index:** Creates indexes for use by the Query service. The Couchbase Index service supports the creation of primary and secondary indexes on items stored within the Couchbase server.
- **Search:** Creates indexes specially purposed for Full Text Search. This supports language-aware searching, allowing users to search for, say, the word *beauties*, and additionally obtain results for *beauty* and *beautiful*.
- **Analytics:** Supports join, set, aggregation, and grouping operations, which are expected to be large, long running, and highly consumptive of memory and CPU resources.
- **Eventing:** Supports near real-time handling of changes to data. Code can be executed both in response to document-mutations, and as scheduled by timers.

- **Backup:** Supports both the scheduling and the immediate execution of full and incremental data backups, either for specific individual buckets, or for all buckets on the cluster. It also allows the scheduling and immediate execution of merges of previously made backups.

Quick installation of Couchbase server using Docker

Let us look at how we can install Couchbase server using Docker and play around with the Couchbase cluster.

To begin, run the following commands on a console window on your computer:

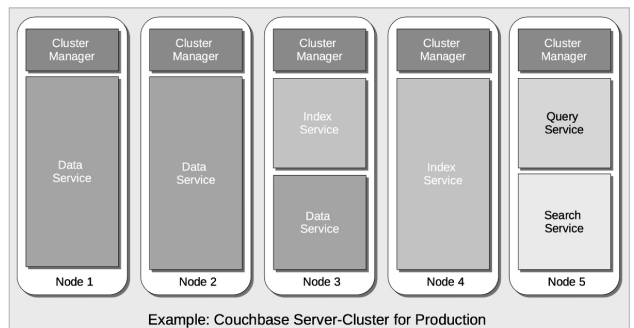


Figure 1: Couchbase server overview

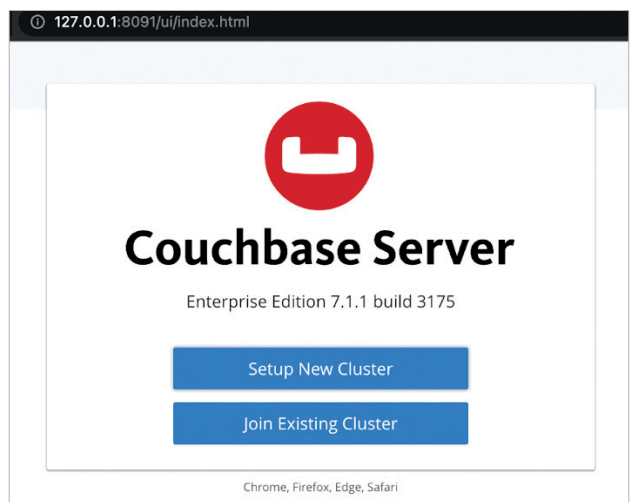


Figure 2: Initial cluster setup page